Network Troubleshooting Assignment

Ping, Traceroute, and NSLOOKUP Tools Overview

Question 1: Test Connectivity to YouTube Using Ping

Task: Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Command:

ping www.youtube.com

Sample Result:

Item	Result
Packets sent	4
Packets received	4
Packet loss	0%
Average response time	89 ms (illustrative example)

Answer: The ping command tests connectivity and measures response time. In this example, the average response time is 89 ms.

Question 2: Run Traceroute to Flipkart

Task: Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Command:
tracert www.flipkart.com

Example of a Possible Route (Illustrative Only):

Hop	Network Device
1	Home Wi-Fi router
2	Internet Service Provider (ISP) gateway
3	ISP network router
4	Regional network router
5	Intermediate internet router
6	CDN or destination network

Answer: Traceroute displays the intermediate hops between my computer and Flipkart. The actual number of hops and IP addresses vary by network, and some hops may not respond.

Question 3: Find Zomato's IP Address Using NSLOOKUP

Task: Use the nslookup tool to find the IP address of www.zomato.com and record the result.

Command:
nslookup www.zomato.com

Example Result:
Public DNS data for zomato.com has included these IPv4 addresses:
• 13.251.252.46
• 18.142.170.124
• 52.220.176.38
The exact result for www.zomato.com may differ from the root domain and can change depending on DNS configuration.

Answer: NSLOOKUP is used to find the IP addresses associated with a domain name and troubleshoot DNS resolution.

Question 4: Ping a Local Network Device

Task: Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Command Example:
ping 192.168.1.10

Answer: If the device is online and permits ping requests, it may respond with its response time. If it is offline or blocks ICMP requests, the output may show Request timed out or Destination host unreachable.

Question 5: Short Report on Network Troubleshooting

Task: Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing specific network problems.

Answer:

1. Ping checks connectivity and response time. For example, ping www.youtube.com tests whether YouTube responds to ping requests.
2. Traceroute identifies the route to a website. For example, tracert www.flipkart.com shows intermediate hops.
3. NSLOOKUP checks DNS resolution. For example, nslookup www.zomato.com returns available IP addresses.
4. For a website that does not load, I would use ping and traceroute to investigate connectivity and routing.
5. For a slow connection, I would compare ping response times and examine traceroute timings for possible delays.
6. For suspected DNS problems, I would use NSLOOKUP to check whether the domain resolves correctly.